

Network Issues and Their Solutions

Below are some prevalent network issues along with their corresponding solutions

–

1. Slow Network Issues

The speed of the network is the measure of the speed of data transfer between a device and a server. We calculate this speed in Mbps or Gbps.

There are various tools through which you can check your network speed. One popular option is [Ookla Speedtest](#).

The slow third-party network slows down the loading of web pages. Files are downloaded at a slow rate. Video calls freeze and stutter. Such poor performance impacts all the devices on your network and makes even the basic tasks painful and time-consuming.

Several reasons may reduce your network speed:

- Too many devices are in the network simultaneously
- Outdated network gear
- Background apps that waste data
- Other electronic interference
- The distance between the router and the rest of the devices
- ISP throttling
- Malware infections

The issues are complicated when these problems appear jointly. Your network may become a mess during peak hours as everyone streams videos or works from the comfort of their homes.

Solution

Network monitoring tools can help figure out what is bottlenecking your connection. These applications provide real-time data use and identify areas of problems. To resolve slow network host problems:

- Install router firmware on the router regularly
- Place your router in a central place
- Move to the 5GHz frequency band
- Lock gadgets when it is crucial

- Make scans of viruses and malware
- Upgrade old network equipment
- Call your ISP about speed problems

2. High Bandwidth Usage

Bandwidth is defined as the rate at which information may be sent through a network like the Internet. It is represented in megabits per second (Mbps).

With higher bandwidth, more devices can connect to your network without slowing down data flow. Congestion occurs in a network when one or more users, or particularly large applications, consume excessive bandwidth by downloading many files at once.

High bandwidth utilization can result in network congestion, which in turn could result in insufficient bandwidth for other sections of your network. When this occurs, you could notice issues with the Internet, such as a decrease in download speed. Some of the reasons that can result in high bandwidth usage are:

- Large downloads
- Latency
- Packet loss
- Video streaming
- Large applications
- File sharing

But the question that arises is how one can reduce high bandwidth usage. Below we have provided the solution for it.

Solution

Network performance monitors (NPMs) can track the amount of bandwidth in use at a given time and pinpoint the exact location of any disruptions in data transfer. Some of the things or steps one can take to reduce the high bandwidth usage are:

- Look for Malware
- Use [Quality of Service \(QoS\)](#)
- Tracking down bandwidth hogs
- Limiting VoIP network usage

3. DNS Issues

DNS, or Domain Name System, is a directory for the internet that simply links domain names with IP addresses. Problems with the [domain name system \(DNS\)](#) are quite widespread yet sometimes go unnoticed. When you have DNS problems, it usually means you no longer have a connection to the Internet or a network.

It's possible that your site seems to be online and running for you while being offline to your visitors simultaneously. DNS issues can be due to many factors, some of these are:

- Bad configurations
- High DNS latency
- High [TTL](#) values
- Hardware or network failures

When customers are unable to reach your website, for example, the loss of Internet connectivity may have a terrible impact on your business. Finding and fixing DNS issues quickly is crucial since even a few hours of downtime may have serious consequences for a business.

Solution

One can rectify the DNS network issues by following the steps that have been discussed below. These are:

- **Flush DNS Cache:** One can flush the DNS cache by using the command “ipconfig /flushdns” in CMD for Windows and “sudo dscacheutil -flushcache; sudo killall -HUP mDNSResponder” in the terminal app for MAC.
- **Updating network drivers or reinstalling:** Sometimes network drivers get corrupted, or newer versions patches might be required for better functioning; hence it is recommended to update network drivers routinely.
- **Change your device's DNS server to a different one:** One can change the device's DNS server such as Google DNS (8.8.8.8 and 8.8.4.4) or OpenDNS (208.67.222.222 and 208.67.220.220).
- **Restart your router:** Reset it to its factory settings and reconfigure it with the correct DNS settings.
- **Changing firewall settings:** Sometimes firewall can act strictly and block some specific DNS addresses. Hence, it is advisable to allow access to the DNS servers or domains that you want to visit on your firewall settings.

4. Packet Loss Issues

Data sent over networks is subdivided into small groups of data called packets. A packet stores a section of your file, email, or video stream. Packet loss occurs when such data packets are unable to reach their destination. This leads to delays in games, stuttering audio in phone conversations and corrupt file downloads. Packet loss of even small amounts leads to noticeable problems.

But, how to track packet loss in your network? You can directly click on [packetlosstest](#) to get an idea of whether you are getting any packet loss or not.

Possible reasons for packet loss are:

- Network congestion
- Improper cables or connections
- Out-fashioned network drivers
- Router overheating
- Wireless interference
- Insufficient bandwidth
- Hardware failures

Solution

Network testing tools measure the percentage of packet loss and identify where the problem is. To have good performance, most networks are expected to have less than 1% packet loss. To minimize loss of packets:

- Replace the damaged Ethernet cable
- Update network adapter drivers
- Reboot your router once a week
- Test the loose connections
- Minimize wireless interference
- Move to business-class hardware
- Monitor network traffic patterns

5. IP Address Conflict

When two or more devices have the same IP address in the same network is known as an IP address conflict. An [IP address](#) is used to communicate and receive data from a specific device or network. Every device has a unique IP address. If two devices have the same IP address, they cannot communicate properly and may

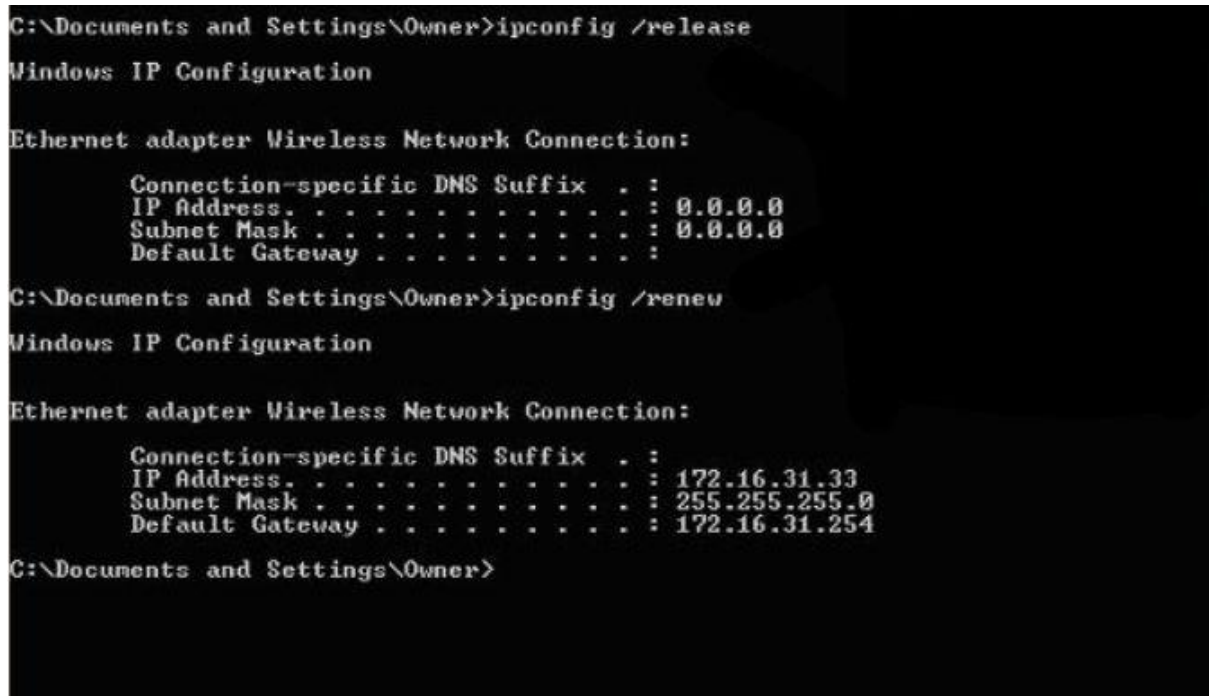
cause network errors or crashes. Some of the reasons that can lead to IP address conflict are:

- A user manually assigns an IP address to a device already used by another device on the network.
- A device obtains an IP address from a [DHCP](#) server already used by another device on the network.
- A device has a static IP address that is not updated when it joins a different network or when the network configuration changes.

Solution

Below we have discussed some of the key steps one can take to remove IP address conflict. These are:

- Release and renew the IP address: Using CMD in Windows and commands “ipconfig /release” and, after that, “ipconfig /renew”.

A screenshot of a Windows Command Prompt window. The prompt shows the user at 'C:\Documents and Settings\Owner>' typing 'ipconfig /release'. The output shows 'Windows IP Configuration' and 'Ethernet adapter Wireless Network Connection:' with fields for 'Connection-specific DNS Suffix', 'IP Address', 'Subnet Mask', and 'Default Gateway', all showing '0.0.0.0'. Then, the user types 'ipconfig /renew'. The output shows the same fields, but with 'IP Address' as '172.16.31.33', 'Subnet Mask' as '255.255.255.0', and 'Default Gateway' as '172.16.31.254'. The prompt ends with 'C:\Documents and Settings\Owner>'.

```
C:\Documents and Settings\Owner>ipconfig /release
Windows IP Configuration

Ethernet adapter Wireless Network Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 0.0.0.0
    Subnet Mask . . . . . : 0.0.0.0
    Default Gateway . . . . . : 

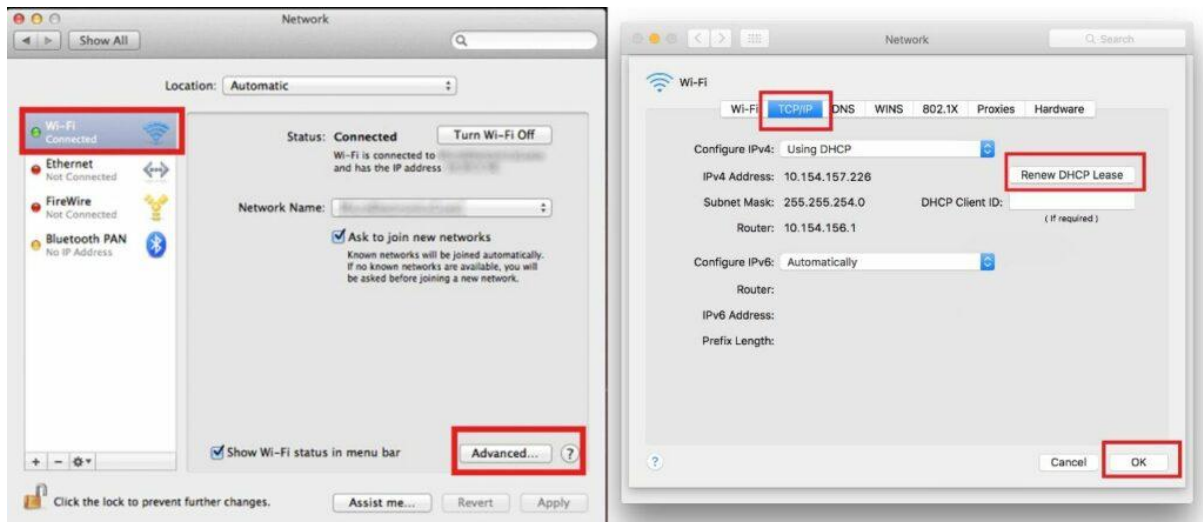
C:\Documents and Settings\Owner>ipconfig /renew
Windows IP Configuration

Ethernet adapter Wireless Network Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 172.16.31.33
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 172.16.31.254

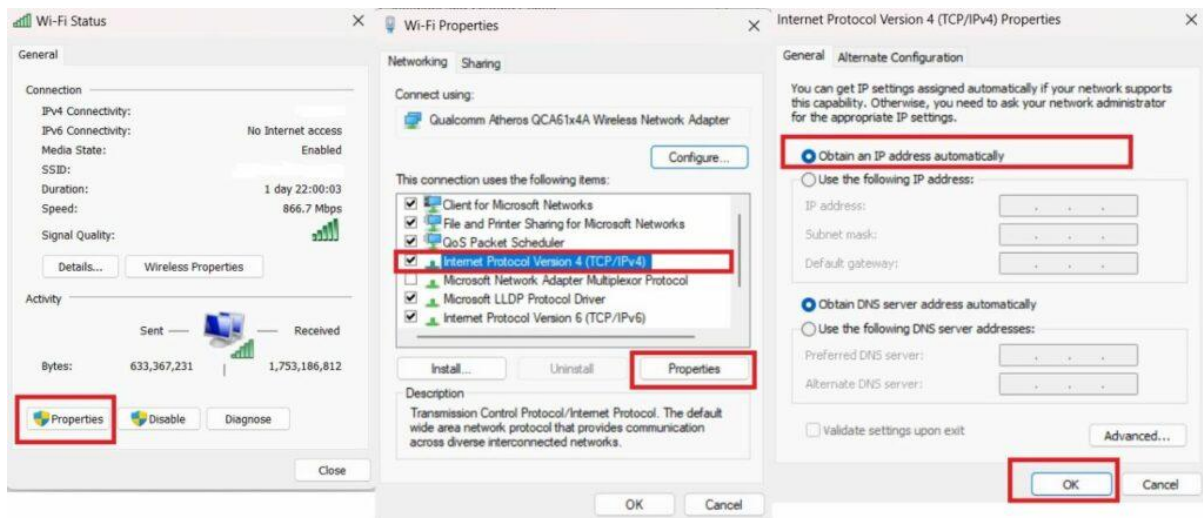
C:\Documents and Settings\Owner>
```

And in the case of Mac, simply go to the “Internet and network section” and select the WIFI or [ethernet](#) you are connected to, after that click on advanced settings, and then go to TCP/IP tab where you have to renew the DHCP lease and go for a system restart afterward. For Mac, we have shown below the picture for easy understanding.



- **Automatic IP address from DHCP server:** Configure your device to obtain an IP address automatically from a DHCP server instead of using a manual or static IP address.

For Windows: Below with the help of the image you can choose an automatic IP address from the DHCP server.



- **Reserving IP address:** Reserve a specific IP address for your device on the router's settings to prevent it from being assigned to another device.

6. Connectivity and Interference Issues in the Wireless Network

The most frequent network issue people have nowadays is with their WIFI. WIFI connectivity issues mean that your device cannot connect to the Wi-Fi network or that the connection is unstable. Wi-Fi connectivity issues can be caused due to many factors, some of which are listed below:

- **Congestion**
- **Distance**

- Interference
- Hardware or Software

Solution

Below we have discussed some of the steps one can take to connect to the wireless network easily. These are:

- **Restarting the router:** Most of the time, several glitches can lead to WIFI connectivity issues, so it's always good to go for restarting the router.
- **Forgetting the WIFI network:** Forget the Wi-Fi network on your device and reconnect to it with the correct password and settings.
- **Changing WIFI frequency and channel:** Sometimes interference can also lead to connectivity issues. Hence, changing your WIFI channel or frequency on your router is advisable.
- **Updating device firmware:** Always update your device's drivers and firmware. Outdated drivers can lead to compatibility or performance issues. This results in WIFI connection issues.

Now, let's talk about the WIFI interference.

7. Wi-Fi Interference

When your Wi-Fi signal from your wireless network is interrupted or weakened, this is known as wireless interference. Some of the signs that show interference in the wireless networks are:

- Even when physically adjacent to a WIFI device (router), there is poor WIFI signal strength.
- WIFI-connected users get a slower Internet connection.
- WIFI-based file transfers are very slow.
- Unable to link a Bluetooth or WIFI device despite being in close proximity to the router.
- The WIFI connection drops every once in a while.

Solution –

WIFI interference can be solved via:

- Repositioning router
- Choosing the right WIFI channel

8. Network Security Issues

Network security issues can lead to many problems by compromising confidentiality, the integrity of a network, or its data. It can lead to a lot of issues, such as:

- Unauthorized access
- Data theft
- Data loss
- Data corruption

Some of the attacks that can lead to network security issues we discussed above are:

- Phishing
- Malware
- DDoS
- Brute force
- Man-in-the-Middle

Solution

It's of utmost importance to protect yourself from such attacks and protect your data as well as your resources. Some of the ways one can rectify network security issues are:

- One can protect their devices and network with various antivirus available, firewalls, and VPN software.
- Keep updating your OS, application drivers as well as security patches that all vendors provide on a monthly basis.

9. High CPU Usage

The Central Processing Unit (CPU) is the core component responsible for receiving and executing program instructions. The likelihood of anything going wrong with your network rises as your devices are asked to do more and more, which further results in lowering the CPU processing role.

The most typical cause of high CPU usage is when network traffic causes bottlenecks. When processes take longer to run or when more network packets are being sent and received in higher amounts than usual, CPU consumption may spike dramatically.

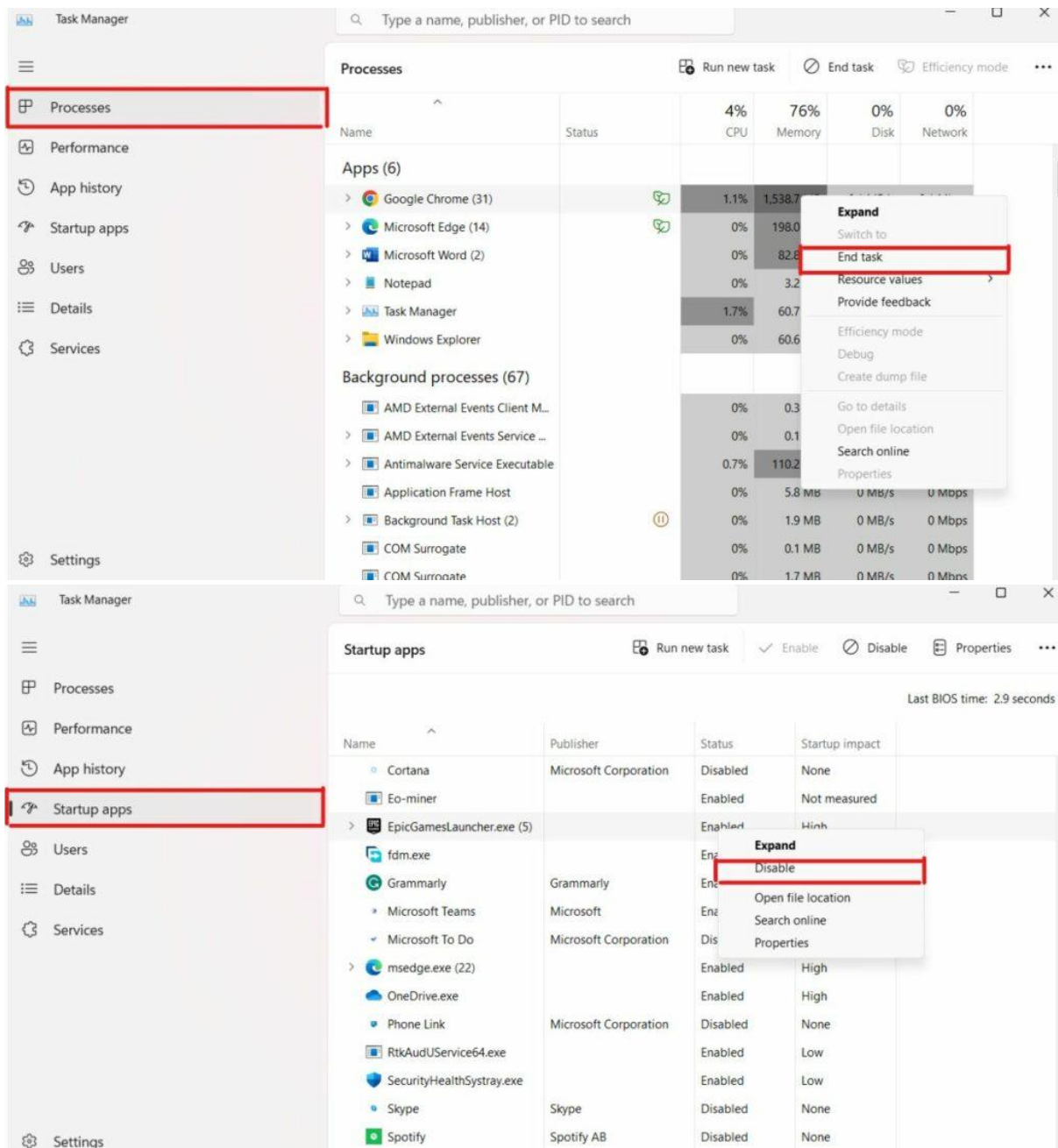
Some of the other reasons that lead to high CPU usage are:

- **Background processes**
- **Virus**
- **System Idle processes**
- **Svchost.exe (netscvs) Process**

Solution

Avoiding CPU-intensive tasks is the best method to keep your machine running smoothly. However, there are situations when it gets out of control due to a glitch in the system. Below we have provided some of the general steps one can take to fix the high CPU usage issue.

- **The first and foremost thing that one must do is find the program or virus that uses the CPU the most. You can do it by opening the task manager ctrl + alt + del. Once you are in the task manager and check the processes section. By this, you will know which app utilizes the CPU the most. Right-click and end the process to free up some CPU usage %. It is highly suggestible to go to the startup apps option and disable the apps you don't often use for fast boot and free up some CPU space.**



- If it's malware or a virus, then always use a highly reputable antivirus to detect and delete the virus that is unnecessarily using the CPU usage.
- If you find that a legitimate program is causing high CPU usage, you may want to update it to the latest version or check for any patches or fixes from the developer. Sometimes, outdated or buggy software can cause compatibility issues or performance problems with your system.

10. Can't Connect to Printer

Network printers give several users the opportunity to print using any device. The failure of the printer connection brings work to a standstill. Sometimes, when the printer is powered on and networked, it appears offline.

Different connection methods are used by modern printers. Some printers are accessed by Wi-Fi, and others use Ethernet cables. There are specific troubleshooting guidelines for every method. Connection problems are usually caused by simple mistakes in setup.

The failure of connection to printers is caused by:

- Wrong IP address configuration
- Printer access blocked by firewall
- Outdated printer drivers
- Invalid network credentials
- Power saving mode problems
- Print spooler service errors
- Network security settings

All these problems increase when there are a number of printers and users within an office environment. A single misconfiguration can prevent entire departments from printing.

Solution

The most common approach to printer troubleshooting is a basic connection. Confirm that the printer is on and ready. Then get down to network-specific solutions:

- Reboot the computer and printer
- Windows printer troubleshooter
- Install the newest printer drivers
- The printer IP address is correct in the settings
- Temporarily turn off the firewall and test
- Clear and restart the print spooler service
- Configure the printer by manually setting IP settings

These are the some of the common computer network issues and their solutions.

Common Causes of Network Problems

Network Problems can arise from various sources, including:

- Hardware faults

- Overloaded Networks
- Software Bugs or misconfigurations
- Physical obstructions and interferences
- Security settings like VPN or firewalls

Basic Network Troubleshooting Steps

When network issues arise, you can use these basic troubleshooting steps:

1. **Checking Physical Connections:** Ensure all cables are correctly connected.
2. Next, you can force restart the routers and other affected devices.
3. Make sure all [network devices](#) have the latest firmware updates.
4. Double-check every setting on each device manually.

Advanced Solutions for Persistent Network Issues

For complex issues that persist, you should consult a networking professional, or you can take these steps:

- Upgrading to enterprise-grade networking hardware
- Implementing Network Segmentation
- Using Load Balancers

Frequently Asked Questions

Q1. What is a network issue?

A network issue is a problem that affects the communication or connectivity between devices or systems on a network. It can be because of many factors, such as physical damage to the [networking equipment](#), viruses, degraded performance, data loss, or service interruptions for the users or applications that rely on the network.

Q2. How do I resolve network issues?

Network issues can be resolved depending on the kind of network issues you are facing. For example, if you are facing difficulty connecting to the internet, you must restart the modem or router or check the drivers to see whether they are updated. More complex issues might require [regular monitoring](#) or replacement of hardware.